1 Social Optimum: Pigou Network

1.1 Network

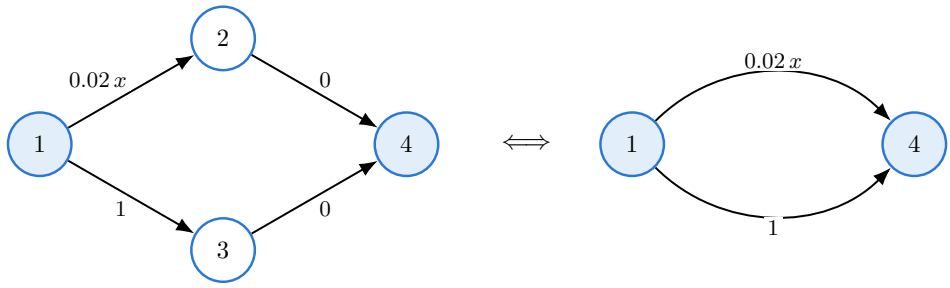


Figure 1: Pigou network. x is the number of agents on an edge in a timestep.

All agents travel $1 \to 4$ on the top route T (cost $0.02x$) or the bottom route B (cost 1). All users of T cross $1 \to 2$ in the same step, so timing plays no role.

1.2 Derivation

With k agents on T, each of them pays $0.02k$, and each of the $N-k$ agents on B pays 1:

$$SC(k) = \underbrace{k \cdot 0.02k}_{T} + \underbrace{(N-k) \cdot 1}_{B} = 0.02k^2 + (N-k).$$

$SC'(k) = 0.04k - 1 = 0$ gives $k = 25$: fill T until its marginal cost reaches B's cost of 1. Since $k \le N$,

$$k^* = \min(N, 25), \qquad SC_{\text{opt}} = \begin{cases} 0.02N^2 & N \le 25 \\ N - 12.5 & N \ge 25 \end{cases}$$

k^* is a whole number, so this is also the optimum for whole agents. Selfish agents instead fill T until it costs 1, so $k_{\text{UE}} = \min(N, 50)$.

1.3 Results

N	On T	On B	SC_{opt}	SC_{opt}/N	SC_{UE}	$SC_{\text{UE}}/SC_{\text{opt}}$
20	20	0	8	0.4	8	1
50	25	25	37.5	0.75	50	4/3
80	25	55	67.5	0.84375	80	$32/27 \approx 1.19$

Table 1: Social optimum and selfish equilibrium. The last column is the price of anarchy.

- $N = 20$: T never gets crowded enough to need B. Selfish routing is already optimal, and any agent that moves to B raises the social cost.
- $N = 50$: with everyone on T it costs exactly 1, the textbook Pigou example, with the worst-case price of anarchy for linear costs (4/3).
- $N > 25$: T keeps 25 agents and the rest pay 1, so $SC_{\text{opt}}/N = 1 - 12.5/N$ rises with N.